

Experiment39:

Develop a C program to simulate C-SCAN disk scheduling algorithm.

Code: -

```
#include <stdio.h>

#include <stdlib.h>

int main()
{
    int n, head, seek_time = 0;

    int disk_size;

    printf("Enter the disk size: ");

    scanf("%d", &disk_size);

    printf("Enter the number of disk requests: ");

    scanf("%d", &n);

    int request_queue[n];

    printf("Enter the disk request queue:\n");

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &request_queue[i]);
    }

    printf("Enter the initial position of the disk head: ");

    scanf("%d", &head);

    for (int i = 0; i < n - 1; i++)
    {
        for (int j = i + 1; j < n; j++)
        {
```

```

        if (request_queue[i] > request_queue[j])
        {
            int temp = request_queue[i];
            request_queue[i] = request_queue[j];
            request_queue[j] = temp;
        }
    }
}

printf("\nC-SCAN Disk Scheduling:\n");
printf("Head Movement Sequence: %d", head);
int index = 0;
while (index < n && request_queue[index] < head)
{
    index++;
}
for (int i = index; i < n; i++)
{
    seek_time += abs(head - request_queue[i]);
    head = request_queue[i];

    printf(" -> %d", head);
}
if (head != disk_size - 1)
{
    seek_time += abs(head - (disk_size - 1));

```

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    head = disk_size - 1;

    printf("-> %d", head);
}
seek_time += disk_size - 1;

head = 0;

printf("-> %d", head);

for (int i = 0; i < index; i++)
{
    seek_time += abs(head - request_queue[i]);

    head = request_queue[i];

    printf("-> %d", head);
}

printf("\nTotal Seek Time: %d\n", seek_time)

printf("Average Seek Time: %.2f\n",

    (float)seek_time / n);

return 0;
}

```

Output:

```

Enter the disk size: 2
Enter the number of disk requests: 1
Enter the disk request queue:
43
Enter the initial position of the disk head: 4

C-SCAN Disk Scheduling:
Head Movement Sequence: 4 -> 43 -> 1 -> 0
Total Seek Time: 82
Average Seek Time: 82.00

...Program finished with exit code 0
Press ENTER to exit console.

```